



# CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)  
REVIEW TEST  
26-06-2021

## JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : ALL STAR BATCHES)

### PAPER-2

Time : 2:45 Hours

Maximum Marks : 210

READ THE INSTRUCTIONS CAREFULLY

#### GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **29** pages and that all the **20** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

#### OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

#### DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take  **$g = 10 \text{ m/s}^2$**  unless otherwise stated.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

## PART-1 : PHYSICS

### SECTION-I(i) : (Maximum Marks: 40)

- This section contains **TEN** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:  
*Full Marks* : +4 If only (all) the correct option(s) is (are) chosen.  
*Partial Marks* : +3 If all the four options are correct but **ONLY** three options are chosen.  
*Partial Marks* : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.  
*Partial Marks* : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.  
*Zero Marks* : 0 If none of the options is chosen (i.e. the question is unanswered).  
*Negative Marks* : -2 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. A coil is in the form of equilateral triangle of side 'a'. The current in the coil is I.

(A) The magnitude of magnetic field at the centroid of the equilateral triangle is  $\frac{3\mu_0 I}{2\pi a}$ .

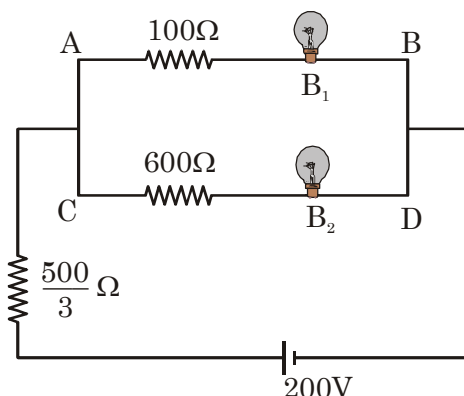
(B) The magnitude of magnetic field at the centroid of equilateral triangle is  $\frac{9\mu_0 I}{2\pi a}$ .

(C) The magnetic field at a distance 'a' equidistant from all the vertices is  $\frac{\mu_0 I}{2\sqrt{3}\pi a}$ .

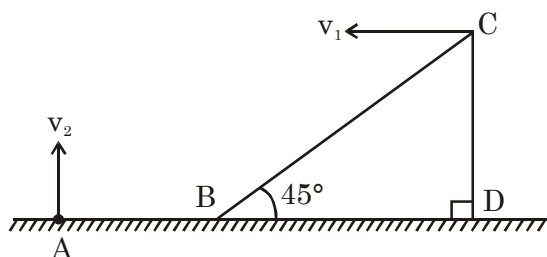
(D) The magnetic field at a distance 'a' equidistant from all the vertices is  $\frac{\mu_0 I}{\sqrt{3}\pi a}$ .

Space for Rough Work

2. Two bulbs ( $B_1 : 25\text{W}, 100\text{ V}$ ) and ( $B_2 : 100\text{ W}, 200\text{ V}$ ) are connected in the circuit as shown in diagram : Choose **INCORRECT** option(s) :



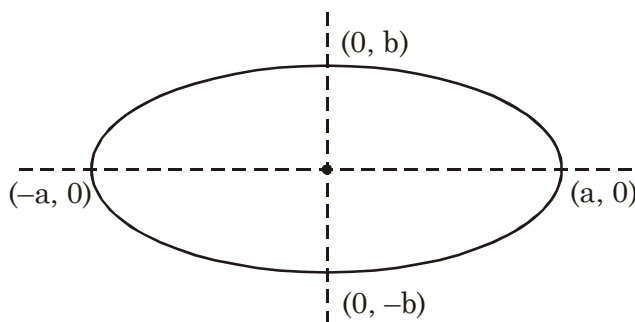
- (A) Heat lost per second in the circuit will be approximately 35 W.  
 (B) Ratio of heat produced per second in both bulbs will be 1 : 1.  
 (C) Ratio of heat produced in branch AB to branch CD will be 1 : 2.  
 (D) Current drawn from the battery is 0.4 A.
3. A large wedge BCD having its inclined surface at an angle  $\theta = 45^\circ$  to the horizontal, is travelling horizontally leftwards with uniform velocity  $v_1 = 10\text{ m/s}$ . At some instant, a particle is projected vertically up with speed  $v_2 = 20\text{ m/s}$  from point A on ground lying at some distance left to the lower edge B of the wedge as shown in figure. The particle strikes the incline BC normally, while it was falling.



- (A) The distance AB at the instant the particle was projected from A, is 15 m.  
 (B) The distance of lower edge B of the wedge from point A at the instant particle strikes the incline, is 15 m.  
 (C) The distance of lower edge B of the wedge from point A at the instant particle strikes the incline, is 30 m.  
 (D) The path of the particle in the reference frame attached to wedge is parabolic.

Space for Rough Work

4. A particle of mass  $m$  is taken with a constant speed  $v_0$  along the elliptical orbit as shown in figure. [semi-major axis length ' $a$ ' and semi minor axis length ' $b$ ']

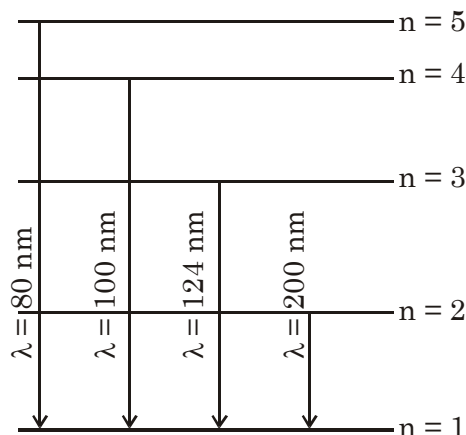


- (A) The centripetal acceleration of particle at  $(0, b)$  is  $\frac{v_0^2 b}{a^2}$ .
- (B) The tangential acceleration of particle at  $(0, b)$  is zero.
- (C) The centripetal acceleration of particle at  $(a, 0)$  is  $\frac{v_0^2 a}{b^2}$ .
- (D) The tangential acceleration of particle at  $(a, 0)$  is  $\frac{v_0^2 a^2}{b^4}$ .

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**Space for Rough Work**

5. In a set of experiments on a hypothetical one-electron atom, the wavelengths of the photons emitted from transitions ending in the ground state are shown in energy level diagram. It takes 17.50 eV to ionize this atom. (Take :  $hc = 1240 \text{ eV}\cdot\text{nm}$ )

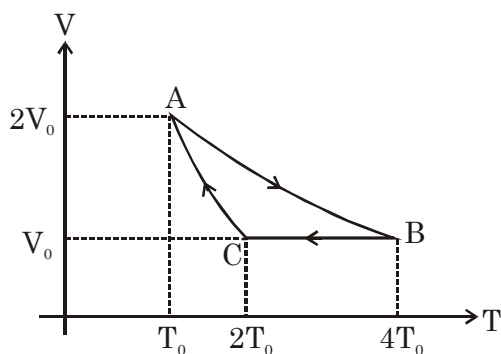


- (A) The energy of the atom in ground state is  $-17.5 \text{ eV}$ .  
 (B) The energy of the atom in  $n = 2$  is  $-11.3 \text{ eV}$ .  
 (C) If the electron makes a transition from  $n = 4$  to  $n = 2$  level, the wavelength of light emitted is  $200 \text{ nm}$ .  
 (D) If the electron makes a transition from  $n = 5$  to  $n = 2$ , the wavelength of light emitted is  $\frac{400}{3} \text{ nm}$ .
6. A siren emits a sinusoidal wave with the frequency  $300 \text{ Hz}$ . The speed of sound is  $340 \text{ m/s}$ . Then the
- (A) wavelength of the waves, if the siren is at rest in the air is  $1.13 \text{ m}$ .  
 (B) wavelength of the wave in front of the source, if the siren is moving at  $40 \text{ m/s}$  is  $1 \text{ m}$ .  
 (C) wavelength of the wave behind the source, if the siren is moving at  $20 \text{ m/s}$ , is  $1.2 \text{ m}$ .  
 (D) wavelength of the wave in front of the stationary source, if the medium is moving with  $20 \text{ m/s}$  towards the observer is  $1 \text{ m}$ .

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Space for Rough Work

7. Figure shows VT diagram for helium gas in a cyclic process. Pressure at point A is  $P_0$ . Then mark the **CORRECT** statement(s).



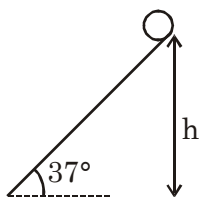
$$AB \Rightarrow TV^2 = \text{constant}$$

$$CA \Rightarrow TV = \text{constant}$$

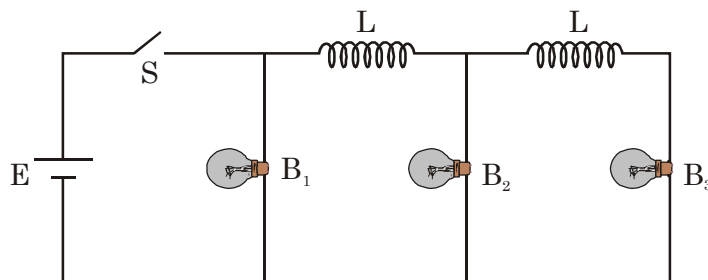
- (A) Molar specific heat capacity of the gas in the process AB is  $R$ .  
 (B) Work done by the gas in the process CA is  $2P_0V_0$ .  
 (C) Net heat given to the system during the process AB is  $6P_0V_0$ .  
 (D) Bulk modulus of the gas during the process CA is  $2P_0$ .

**Space for Rough Work**

8. A small disc of radius  $r$  is released from the top of a rough inclined plane as shown. The coefficient of friction between disc and plane is  $\mu$ . Choose the correct statement(s). Assume that while moving down, the plane of disc always remains vertical.



- (A) If  $\mu = 0$ , no heat is dissipated  
 (B) If  $\mu = 0.4$ , no heat is dissipated  
 (C) if  $\mu = \frac{3}{4} \left( 1 - \sqrt{\frac{2}{3}} \right)$ , maximum heat is dissipated  
 (D) If  $\mu = \frac{3}{4} (2 - \sqrt{2})$ , maximum heat is dissipated
9. In the circuit shown, switch  $S$  is closed for a very long time and each bulb is of equal power rating. Both the inductor coil are of equal inductance. Just after the switch is opened :-



- (A) All the bulbs remain equally bright at all times.  
 (B) Brightness of bulb  $B_1$  becomes 4 times its original brightness.  
 (C) The brightness of bulb  $B_1$  is zero after the switch is opened.  
 (D) Bulb  $B_2$  and bulb  $B_3$  always remain equally bright.

Space for Rough Work

10. The diameter of a wire of length 50 cm is measured with the help of a screw gauge. The main scale reading is 1mm and circular scale reading is 25. Pitch of the screw gauge is 1mm and the total number of divisions on the circular scale is 100. This wire is used in an experiment of determination for the Young's modulus of material of a wire by Searle's method. The wire is placed under a tension of 50 N and elongation of wire is  $\ell = 0.125$  cm. The least count for measuring the length of wire is 0.01 cm and of elongation of the wire is 0.001 cm.
- (A) The least count in the measurement of diameter of wire is 0.01 mm.
- (B) The diameter of the wire is  $(1.25 \pm 0.01)$  mm.
- (C) The percentage error in calculation of Young's modulus is 2.4%.
- (D) The Young's modulus of wire is  $(5.0 \pm 0.2) \times 10^{10}$  N/m<sup>2</sup>.

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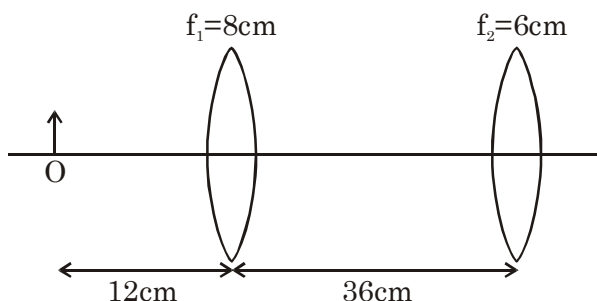
**Space for Rough Work**

**SECTION-I(ii) : (Maximum Marks : 18)**

- This section contains **THREE** paragraphs.
- Based on each paragraph, there are **TWO** questions.
- Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :  
*Full Marks* : +3 If only the bubble corresponding to the correct answer is darkened.  
*Zero Marks* : 0 If none of the bubbles is darkened.  
*Negative Marks* : -1 In all other cases

**Paragraph for Questions 11 and 12**

An object 8 cm high is placed 12 cm to the left of a converging lens of focal length 8 cm. A second converging lens of focal length 6 cm is placed 36 cm to the right of the first lens. Both lens has the same optic axis.

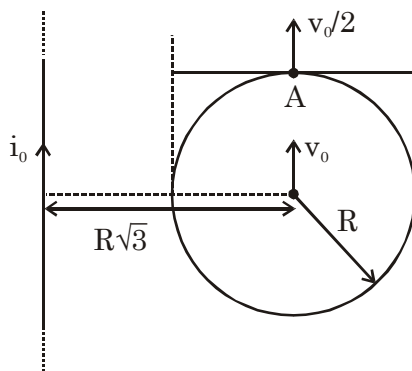


11. The final image formed by the combination of two lens is :  
(A) Real, erect and magnified. (B) Real, inverted and diminished.  
(C) Virtual, erect and magnified. (D) Virtual, inverted and diminished.
12. If the side of second equiconvex lens which is not facing first lens is silvered and the refractive index of material of second lens is 1.5, the image formed after reflection from second lens is  
(A) 1.5 cm to the left of silvered lens. (B)  $\frac{12}{7}$  cm to the left of silvered lens.  
(C) 1.5 cm to the right of silvered lens. (D)  $\frac{12}{7}$  cm to the right of silvered lens.

**Space for Rough Work**

**Paragraph for Questions 13 and 14**

A circular loop of radius  $R$  and resistance per unit length  $\lambda$  is moving with constant velocity  $v_0$  parallel to a very long current carrying wire carrying current  $i_0$ . A resistanceless rod of length  $2R$  is also moving parallel to the wire as shown in the figure. The rod and the loop are in contact at point A and rod is symmetrical with respect to the center of the loop and at  $t = 0$ , rod starts sliding over the loop without friction. Neglect self inductance.



13. The current through the rod when it is at a distance  $\left(\frac{R}{2}\right)$  from the point A of the loop will be:

(A)  $\frac{v_0 \mu_0 i_0}{16 R \lambda \pi^2} \ln 3$  (B)  $\frac{16 v_0 \mu_0 i_0}{R \lambda \pi^2} \ln 3$  (C)  $\frac{9 v_0 \mu_0 i_0}{16 R \lambda \pi^2} \ln 3$  (D) Zero

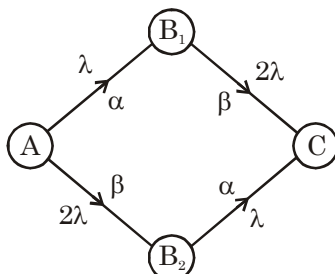
14. Force required to maintain the uniform velocity of the rod at this instant is :

(A)  $\frac{9 \mu_0^2 i_0^2 v_0 (\ln 3)^2}{16 R \lambda \pi^3}$  (B)  $\frac{9 \mu_0^2 i_0^2 v_0 (\ln 3)^2}{R \lambda \pi^3}$  (C)  $\frac{\mu_0^2 i_0^2 v_0 (\ln 3)^2}{32 R \lambda \pi^3}$  (D)  $\frac{9 \mu_0^2 i_0^2 v_0 (\ln 3)^2}{32 R \lambda \pi^3}$

**Space for Rough Work**

**Paragraph for Questions 15 and 16**

A hypothetical decay chain consists of the following elements; A, B<sub>1</sub>, B<sub>2</sub>, and C, where C is stable. The decay constants are  $\lambda$ ,  $2\lambda$  for  $\alpha$  and  $\beta$  decay of A respectively and  $2\lambda$ ,  $\lambda$  for  $\beta$ ,  $\alpha$  decays for B<sub>1</sub> and B<sub>2</sub> respectively leading to the formation of C. Initially number of atoms of A is N<sub>0</sub>.



15. If at some instant, the number of atoms of B<sub>1</sub> reaches a maximum value, the ratio of the number of atoms A to that B<sub>1</sub> at this instant is :
- (A)  $\frac{1}{2}$                       (B) 2                      (C) 4                      (D)  $\frac{1}{4}$
16. The number of atoms of A as a function of time is
- (A)  $N_0 e^{-5\lambda t}$                       (B)  $N_0 e^{-\lambda t}$                       (C)  $N_0 e^{-2\lambda t}$                       (D)  $N_0 e^{-3\lambda t}$

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**Space for Rough Work**

SECTION-II : (Maximum Marks: 12)

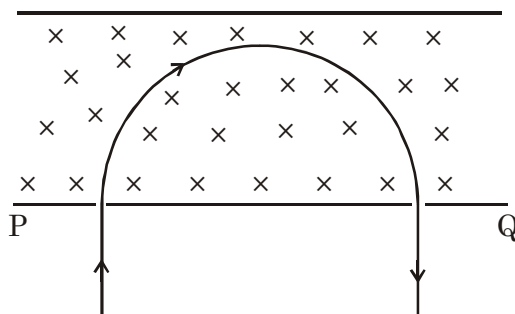
- This section contains **FOUR** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

**For Example :** If answer is -77.25, 5.2 then fill the bubbles as follows.

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| + |   |   |   |   |   |   |   |   |   | - |   |   |   |   |   |   |   |   |   |
| ● |   |   |   |   | ● |   |   |   |   | ● |   |   |   |   | ● |   |   |   |   |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 |
| 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 |

- Answer to each question will be evaluated according to the following marking scheme:  
*Full Marks* : +3 If **ONLY** the correct numerical value is entered as answer.  
*Zero Marks* : 0 In all other cases.

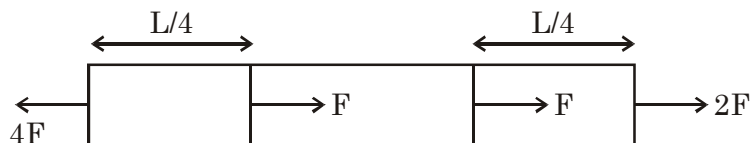
1. A uniform magnetic field with a slit system as shown in the figure is to be used as a velocity filter for high energy charged particles (enter and exit perpendicular to PQ) with a field of B tesla. It is found that the filter transmits  $\alpha$  particle each of energy 2.2 MeV. The magnetic field is increased to 2.13 B Tesla and deuteron ions are passed into the filter. The energy (in MeV) of each deuteron ions transmitted by the filter, is :



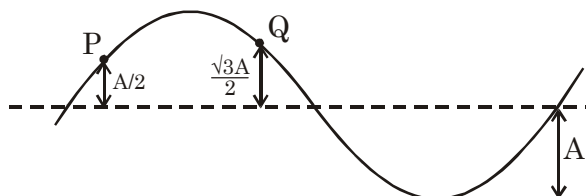
Space for Rough Work

**ALLEN**

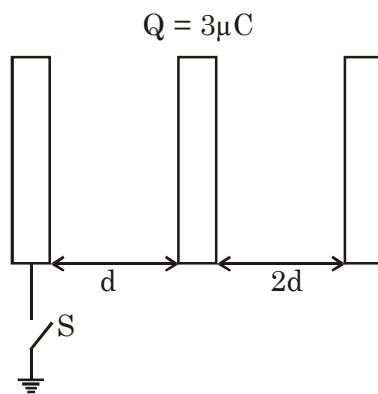
2. A uniform bar of length 'L' and cross-sectional 'A' and Young's modulus 'Y' is acted upon by the forces as shown in figure. The elongation in the rod is  $n \frac{FL}{AY}$ . The value of n is



3. Figure shows the snapshot of the transverse sinusoidal wave on a string at  $t = 0$  sec. Frequency of wave is  $\frac{125}{6}$  Hz.  $T_1$  is the time at which the line joining P and Q become horizontal for first time after  $t = 0$  if wave is travelling towards +ve x-direction and  $T_2$  is the time at which line P and Q become horizontal for the first time after  $t = 0$  if wave is travelling in -ve x-direction. Then the value of  $\frac{T_1}{T_2}$  is



4. The modulus of charge (in  $\mu\text{C}$ ) flown from the earth to the plate after the switch is closed, is




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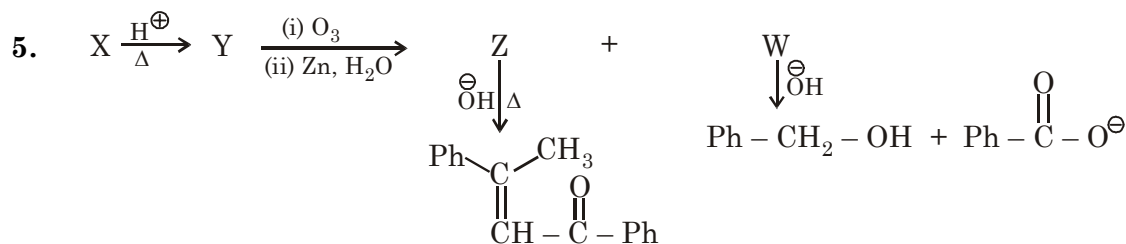
Space for Rough Work

**PART-2 : CHEMISTRY****SECTION-I(i) : (Maximum Marks: 40)**

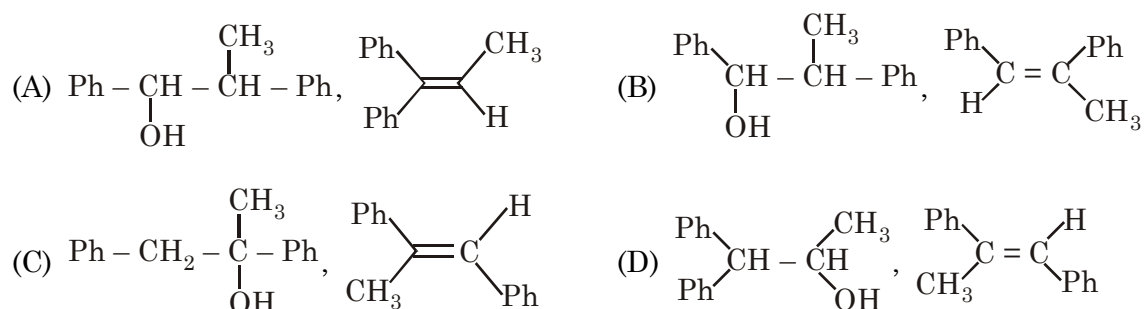
- This section contains **TEN** questions.
  - Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
  - For each question, choose the correct option(s) to answer the question.
  - Answer to each question will be evaluated according to the following marking scheme:  
*Full Marks* : +4 If only (all) the correct option(s) is (are) chosen.  
*Partial Marks* : +3 If all the four options are correct but **ONLY** three options are chosen.  
*Partial Marks* : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.  
*Partial Marks* : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.  
*Zero Marks* : 0 If none of the options is chosen (i.e. the question is unanswered).  
*Negative Marks* : -2 In all other cases.
  - **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.
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1. 400 ml of 0.1M HCl, 500 ml of 0.3 M  $\text{H}_2\text{SO}_4$  & 100 ml of 0.1 M  $\text{KNO}_3$  are mixed then what is correct for resultant solution ?  
(A)  $[\text{H}^+] = 0.19 \text{ M}$  (B)  $[\text{Cl}^-] = 0.04 \text{ M}$  (C)  $[\text{NO}_3^-] = 0.01 \text{ M}$  (D)  $[\text{K}^+] = 0.01 \text{ M}$
  2. What is the equivalent wt. of adenine ( $\text{C}_5\text{H}_5\text{N}_5$ ) when it undergoes complete combustion with excess air ? [M = Molar mass]  
(A)  $\frac{M}{20}$  (B)  $\frac{M}{12}$  (C)  $\frac{M}{25}$  (D)  $\frac{M}{2}$
  3. Select correct statement for a vanderwaals gas ?  
(A) Volume correction can be neglected at low pressure conditions  
(B) Volume correction will be significant at low pressure conditions  
(C) 'b' is four times the volume of one mole of gas molecules  
(D) At very low pressure, ideal behaviour is observed.
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**Space for Rough Work**

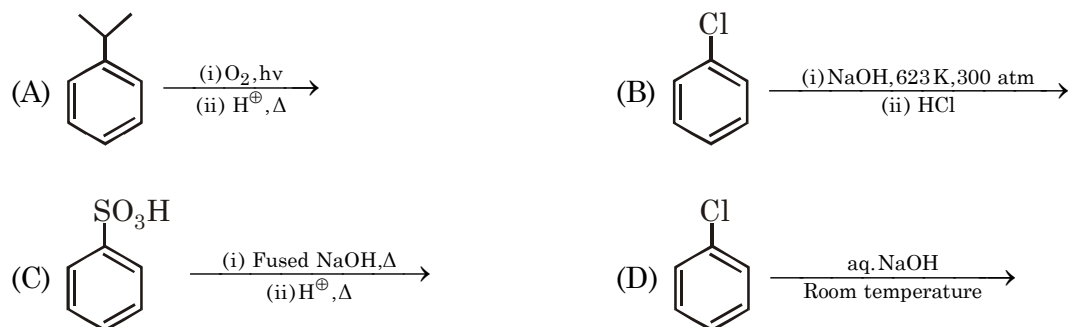
4. Relation between D-glucose and L-glucose is/are :  
 (A) They are C-5 epimers (B) They are enantiomers  
 (C) They are diastereomers (D) They are stereoisomers



X and Y respectively can be:

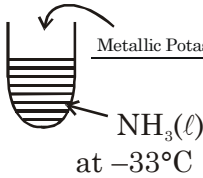


6. Which of the following reaction gives phenol as major product



Space for Rough Work

7.  $\text{Cl}_2 + \text{dry slaked lime} \longrightarrow \text{Product}$   
 Select **CORRECT** about above reaction & product of reaction.  
 (A) It is disproportionation reaction  
 (B) Product gives brick red colour on flame  
 (C) Product can decolorize coloured organic matter  
 (D) Oxidation number of chlorine is +1 & +5 in product
8. Ionization energy of N-family elements (N, P, As)  
 (A) Decrease on moving  $\text{N} \rightarrow \text{P} \rightarrow \text{As}$   
 (B) Higher than that of group 14 elements in the corresponding periods  
 (C) Lower than that of group 16 elements in the corresponding periods  
 (D) The order of successive ionisation energy  $\text{IE}_1 < \text{IE}_2 < \text{IE}_3$  for each atom
9. Which reaction does **NOT** show oxidising action of  $\text{H}_2\text{O}_2$ ?  
 (A)  $[\text{Fe}(\text{CN})_6]^{4-} + \text{H}_2\text{O}_2/\text{H}^+ \longrightarrow [\text{Fe}(\text{CN})_6]^{3-} + \text{other products}$   
 (B)  $[\text{MnO}_4]^- + \text{H}_2\text{O}_2/\text{H}^+ \longrightarrow [\text{Mn}(\text{H}_2\text{O})_6]^{+2} + \text{other products}$   
 (C)  $[\text{MnO}_4]^- + \text{H}_2\text{O}_2/\text{OH}^- \longrightarrow \text{MnO}_2 + \text{other products}$   
 (D)  $[\text{Cr}(\text{OH})_4]^- + \text{H}_2\text{O}_2/\text{OH}^- \longrightarrow \text{CrO}_4^{2-} + \text{other products}$

10.   $\xrightarrow{\text{Metallic Potassium}}$  Product solution

Select **CORRECT** for product solution :-

- (A) Dilute solution is dark blue in colour  
 (B) Similar colour is obtained on adding Na metal instead of potassium  
 (C) Metal can be recovered by vaporization of  $\text{NH}_3$   
 (D) It contain amoniated metal ion & amoniated electron

Space for Rough Work

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**SECTION-I(ii) : (Maximum Marks : 18)**

- This section contains **THREE** paragraphs.
  - Based on each paragraph, there are **TWO** questions.
  - Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four options is correct.
  - For each question, darken the bubble corresponding to the correct option in the ORS.
  - For each question, marks will be awarded in one of the following categories :  
*Full Marks* : +3 If only the bubble corresponding to the correct answer is darkened.  
*Zero Marks* : 0 If none of the bubbles is darkened.  
*Negative Marks* : -1 In all other cases
- 

**Paragraph for Questions 11 and 12**

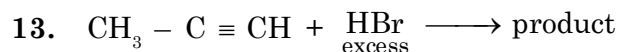
1 litre of  $\text{Na}_2\text{SO}_4$  (aq.) is electrolysed by passing 96.5 amp current for 100 sec. Current efficiency is 80% in electrolysis.

11. Total moles of the gases produced in electrolysis -  
(A) 0.01 (B) 0.02 (C) 0.03 (D) 0.06
12. Total work done by the gases produced, if expanded 32 times the initial volume at  $27^\circ\text{C}$  in reversible adiabatic process ?  
(A) 10.125 R (B) 11.25 R (C) 20.25 R (D) 33.75 R
- 

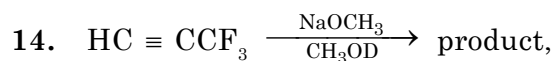
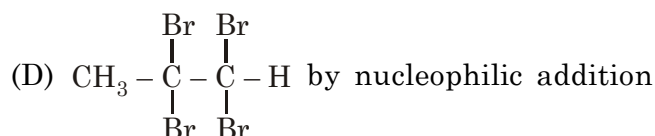
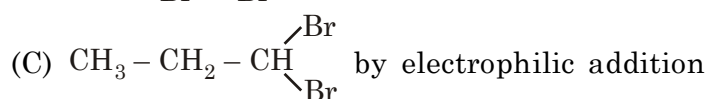
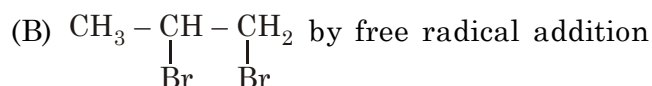
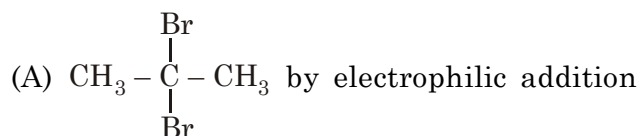
**Space for Rough Work**

Paragraph for Questions 13 and 14

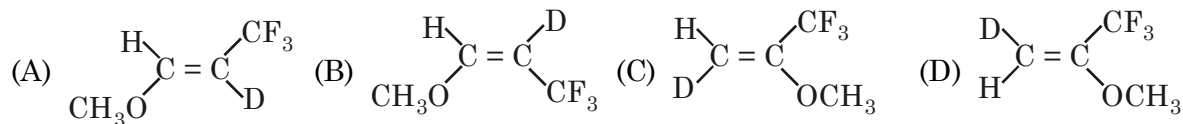
In general electrophilic addition takes place with alkyne in presence of strong electrophile on the other hand nucleophilic addition can occur with alkyne that bear strong electron-attracting substituent such as  $-\text{CF}_3$  on the triple bond and the stereo of addition to triple bond is anti :



Product is :



product is



Space for Rough Work

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**Paragraph for Questions 15 and 16**

In Pyro metallurgical extraction of metal, thermal energy is one of the important inputs. Different unit processes like roasting, smelting, converting and refining require large amount of thermal energy. Roasting is the oxidation of metal sulphides to give metal oxides and sulphur dioxide. In addition other reactions may take place formation of  $\text{SO}_3$  and metal sulphates and formation of complex oxides such as  $\text{ZnO} \cdot \text{Fe}_2\text{O}_3$ .

Typically, sulphide ores of copper, zinc and lead are roasted either partially or completely.

15. "Roasting of ore followed by reduction with coke" is the common method of the extraction of .

(A) Zinc from Calamine (B) Zinc from Sphalerite  
(C) Silver from Argentite (D) Calcium from Dolomite

16. \_\_\_\_\_ is charged in to silica lined convertor, and hot air blast is blown to convert the \_\_\_\_\_ to metallic form, solidified product obtained by this method is called\_\_\_\_\_.

Select **CORRECT** option to fill the blank space repectively to make the above statement true.

(A) Blister copper ;  $\text{Cu}_2\text{S}$  &  $\text{Cu}_2\text{O}$  ; Copper matte  
(B)  $\text{Cu}_2\text{S}$  &  $\text{Cu}_2\text{O}$  ; Blister copper ; Copper matte  
(C) Azurite ;  $\text{Cu}_2\text{S}$  &  $\text{Cu}_2\text{O}$  ; Copper Matte  
(D) Copper matte ;  $\text{Cu}_2\text{S}$  &  $\text{Cu}_2\text{O}$  ; Blister copper

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**Space for Rough Work**

**ALLEN****SECTION-II : (Maximum Marks: 12)**

- This section contains **FOUR** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

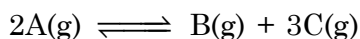
**For Example :** If answer is -77.25, 5.2 then fill the bubbles as follows.

|   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|
|   |   |   |   |   |   |   |   |   |   | + |   |
|   |   |   |   |   |   |   |   |   |   |   |   |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 |
| 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 |

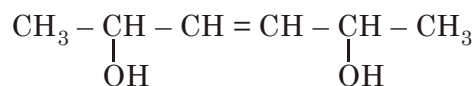
|   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|
|   |   |   |   |   |   |   |   |   |   | - |   |
|   |   |   |   |   |   |   |   |   |   |   |   |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 |
| 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 |

- Answer to each question will be evaluated according to the following marking scheme:  
*Full Marks* : +3 If **ONLY** the correct numerical value is entered as answer.  
*Zero Marks* : 0 In all other cases.

- One mole of A(g) is decomposed to B(g) & C(g) at 27°C in 0.2 L closed vessel. If vapour density observed at equilibrium is 25. Then find  $K_c$  for the reaction : (Mw. of A = 70)



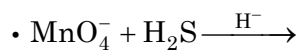
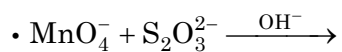
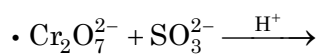
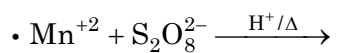
- How many total optically active stereo isomers are possible with following structure?




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**Space for Rough Work**

3. How many enantiomeric pairs are formed on monochlorination of 3-methylpentane?
4. Find sum of the oxidation number of sulphur containing product in all the given reactions



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Space for Rough Work

**PART-3 : MATHEMATICS****SECTION-I(i) : (Maximum Marks: 40)**

- This section contains **TEN** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:  
*Full Marks* : +4 If only (all) the correct option(s) is (are) chosen.  
*Partial Marks* : +3 If all the four options are correct but **ONLY** three options are chosen.  
*Partial Marks* : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.  
*Partial Marks* : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.  
*Zero Marks* : 0 If none of the options is chosen (i.e. the question is unanswered).  
*Negative Marks* : -2 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Let  $f : \mathbb{R} \rightarrow \mathbb{R}$ , be a continuous function satisfying  $f(2x) = f(x) \forall x \in \mathbb{R}$  &  $f(2) = 1$ , then

(A)  $f(3) = 1$

(B)  $y = f(x)$  is many one-into

(C)  $y = f(x)$  is one-one into

(D)  $\int_{-2}^2 f(f(x)) dx = 4$

2. If  $f : (-1, \infty) \rightarrow \mathbb{R}$  is given by  $f(x) = \int \frac{xe^x}{(x+1)^2} dx$  &  $f(0) = 1$ , then

(A)  $y = f(x)$  is non-monotonic but positive in its domain

(B)  $f\left(-\frac{1}{2}\right) > f\left(-\frac{1}{3}\right)$

(C)  $[f(1)] = 1$

(D)  $\int_0^1 f(x) dx > 1$

(where  $[.]$ , denote greatest integer function)

**Space for Rough Work**

3. Lines  $y = m_1x + 2$  &  $y = m_2x + 3$  intersect x-axis at P, Q, respectively & y-axis at R, S respectively. If P, Q, R, S are concyclic and circle passes through P, Q, R and S is  $x^2 + y^2 + 2gx + 2fy + c = 0$ , then
- (A) Inclination of lines from positive direction of x-axis are complementary  
 (B) Value of  $c = 6$   
 (C) Value of  $|c| = 1$   
 (D) If  $m_1 = 2$ , then length of intercept made by circle on x-axis is 5
4. Number of strictly increasing harmonic progressions a, b, c (a, b, c are integers) in which a = 10 & b divides c is
- (A) 2 (B) 3 (C) 4 (D) 5
5. Consider the ellipse  $\frac{x^2}{25} + \frac{y^2}{16} = 1$ . Extremity of its latus-rectum are  $L_1, L_2, L_3$  &  $L_4$  in I<sup>st</sup>, II<sup>nd</sup>, III<sup>rd</sup> & IV<sup>th</sup> quadrant respectively. A concave up parabola having ends of its latus rectum are  $L_1$  &  $L_2$ . Then choose correct option(s).
- (A) length of latus rectum of parabola = 6  
 (B) coordinate of vertex of parabola is  $\left(0, \frac{17}{10}\right)$   
 (C) equation of directrix of parabola is  $y = -\frac{1}{5}$   
 (D)  $\frac{x^2}{5} - \frac{y^2}{4} = 1$  is orthogonal to given ellipse

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Space for Rough Work

6. 5 boys & 5 girls sit in a row randomly, then

(A) Probability that 5 girls sit together =  $\frac{5!6}{10!}$

(B) Probability that boys & girls sit alternatively =  $\frac{5!5}{10!}$

(C) Probability that all boys are together & all girls are together given that all girls are together =  $\frac{1}{3}$

(D) A particular boy & girl sit together given that boys & girls sit alternatively =  $\frac{9}{25}$

7. Consider equation  $z^8 + z^7 + z^6 + \dots + z^2 + z + 1 = 0$  has a non-real root  $\alpha$  (having least positive argument), then choose correct option(s)

(A)  $\prod_{i=1}^8 (2 - \alpha^i) = 511$

(B)  $\sum_{i=1}^4 \operatorname{Re}(\alpha^i) = -\frac{1}{2}$

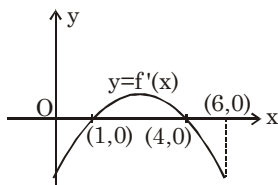
(C)  $\frac{\sum_{i=0}^8 |\alpha^i - \alpha^{i+1}|}{\sum_{i=0}^4 |\alpha^{2i} - \alpha^{2i+1}|} = 2$

(D)  $\sum_{i=1}^4 \alpha^i \alpha^{9-i} = 4$

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Space for Rough Work

8.  $y = f(x)$  is a polynomial function passes through (1, 2), (4, 6) & (6, 0). If graph of  $y = f'(x)$  is shown below



then choose correct option(s)

- (A)  $f(x) = 0$  has no solution in (0, 6)  
 (B)  $y = f(x)$  will have one point of maxima & one minima in (0, 6)  
 (C)  $6 < \int_1^4 f(x) dx < 18$   
 (D) Value of  $[f(1^+) - 1] + [7 - f(4^-)]$  is equal to 2

Here  $[.]$  denotes greatest integer function

9. Let  $y = f(x)$  be a differentiable function  $\forall x \in \mathbb{R}$  & satisfy  $f(x) = x + \int_0^1 x^2 f(t) dt$ . Then choose

correct option(s) related to  $y = f(x)$

- (A) minimum value of  $f(x)$  is  $-\frac{1}{3}$       (B)  $\cos^{-1}\left(f\left(-\frac{4}{3}\right)\right) + \cot^{-1}(f(0))$  will be equal to  $\frac{\pi}{2}$   
 (C)  $\lim_{x \rightarrow \infty} f(x) \cdot f\left(\frac{1}{x^2}\right) = \frac{3}{4}$       (D) Area bounded by curve  $y = f(x)$ , x-axis is  $\frac{8}{27}$

Space for Rough Work

10. Consider  $S_1 : x^2 + y^2 - 1 = 0$   
 $S_2 : (x - 8)^2 + (y - 0)^2 = 25$   
 $S_3 : (x - 7)^2 + (y - 1)^2 = 1$

locus of centre of variable circle touches both  $S_1$  &  $S_2$  is conic  $C_1$

locus of centre of variable circle touches both  $S_2$  &  $S_3$  is conic  $C_2$

locus of centre of variable circle touches both  $S_1$  & y-axis is conic  $C_3$

then choose correct option(s) :-

(A) eccentricity of conic  $C_1$  is 2

(B) eccentricity of conic  $C_2$  is  $\frac{1}{3\sqrt{2}}$

(C) eccentricity of conic  $C_3$  is 1

(D) diameter of largest circle touches both  $S_2$  &  $S_3$  is  $6 + \sqrt{2}$

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**Space for Rough Work**

**SECTION-I(ii) : (Maximum Marks : 18)**

- This section contains **THREE** paragraphs.
- Based on each paragraph, there are **TWO** questions.
- Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :  
*Full Marks* : +3 If only the bubble corresponding to the correct answer is darkened.  
*Zero Marks* : 0 If none of the bubbles is darkened.  
*Negative Marks* : -1 In all other cases

**Paragraph for Questions 11 and 12**

Let  $L_1 : x = y = z$

$L_2 : x + 1 = y + 2 = z + 3$  be two lines

From origin O on  $L_1$ , perpendicular drawn to  $L_2$  has its foot at M. Segment OM is rotated about origin O by an angle  $90^\circ$  &  $180^\circ$  such that  $L_2$  moves alongwith it, without changing its direction & hence  $L_2$  becomes  $L_3$  &  $L_4$  respectively after rotation. In whole process M becomes N & P while rotating  $90^\circ$  &  $180^\circ$  respectively.

On the basis of above information, answer the following questions :

11. Direction ratio of line ON is  
 (A)  $\langle 1, 2, 1 \rangle$  (B)  $\langle -1, 2, 1 \rangle$  (C)  $\langle 1, -2, 1 \rangle$  (D)  $\langle -1, -2, -1 \rangle$
12. Coordinate of point P is  
 (A)  $(1, 0, -1)$  (B)  $(1, -1, 0)$  (C)  $(0, -1, 1)$  (D)  $(-1, 0, 1)$

**Space for Rough Work**

**Paragraph for Questions 13 and 14**

Let set A contain all the  $3 \times 3$  matrices whose entries are 0, 1 & -1. Two of these entries are 1, two are -1 & five are 0.

On the basis of above information, answer the following questions :

13. Number of skew-symmetric matrix in set A is  
(A) 2 (B) 4 (C) 6 (D) 12
14. Number of symmetric matrix in set A which is non-singular is  
(A) 4 (B) 6 (C) 12 (D) 16

**Paragraph for Questions 15 and 16**

A differentiable function  $y = f(x)$  satisfy  $\int_0^x (x-t+1)f(t) dt = x^4 + x^2 \forall x \geq 0$

On the basis of above information, answer the following questions :

15.  $f(1)$  lie in the interval  
(A) (1, 3) (B) (6, 8) (C) (4, 6) (D) (8, 10)
16. If  $f(x) + f'(x) = p(x)$ , then area bounded by  $y = p^{-1}(x)$ , x-axis &  $x = 14$  is equal to  
(A) 2 (B) 4 (C) 8 (D) 12

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**Space for Rough Work**

SECTION-II : (Maximum Marks: 12)

- This section contains **FOUR** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

**For Example :** If answer is -77.25, 5.2 then fill the bubbles as follows.

|   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|
|   |   |   |   |   |   |   |   |   |   | + |   |
|   |   |   |   |   |   |   |   |   |   |   |   |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 |
| 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 |

|   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|
|   |   |   |   |   |   |   |   |   |   | - |   |
|   |   |   |   |   |   |   |   |   |   |   |   |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 |
| 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 |

- Answer to each question will be evaluated according to the following marking scheme:  
*Full Marks* : +3 If **ONLY** the correct numerical value is entered as answer.  
*Zero Marks* : 0 In all other cases.

1. If number of straight line formed by 10 points (no three of them are collinear) =  $\sum_{r=0}^{m-3} {}^m C_r \cdot {}^m C_{r+3}$   
then value of  $\frac{1}{m}$  is equal to
2. If equations  $x^2 + ax + b = 0$  ( $a, b \in \mathbb{R}$ ) &  $x^3 + 3x^2 + 5x + 3 = 0$  have two common roots, then value of  $\frac{b}{a}$  is equal to

Space for Rough Work

3. Let  $\vec{u}, \vec{v}, \vec{w}$  be three unit vectors such that  $\vec{u} \times \vec{v} = \vec{w}$ . If  $\vec{a}$  is any vector such that  $[\vec{a} \vec{v} \vec{w}] = 2$ ,  $[\vec{a} \vec{w} \vec{u}] = 3$  &  $[\vec{a} \vec{u} \vec{v}] = 4$ , then value of  $\frac{\text{projection of } \vec{a} \text{ upon } \vec{w}}{\text{projection of } \vec{a} \text{ upon } \vec{v}}$
4. Three circle touches one another externally. The radius of circles are three consecutive integers. The tangent at their point of contact meet at a point whose distance from a point of contact is 4. The ratio of radius of largest to smallest circle is

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**Space for Rough Work**